SQL for Data Science

Assignment 1

Assignment Information

Assessment Title :	Developing a Power BI Dashboard for Olympic Games Analysis		
Programme Code(s):	NA	**Issue date[1]:**	23/07/2024
Weighting :	15% of the total module mark	**Submission date[2]:**	11/08/2024
Assessor(s) :	Dr Charith Silva & Ms Chamodhi Wickramasinghe	**Return date[3]:**	The final mark will be available in a maximum of 4 weeks after submission.

[1] Date on which brief was given to students

Learning Outcomes of this Assessment

The learning outcomes covered by this assignment are you will be able to, provide a broad overview of the general field of visualisation and dashboard designing.

Key Skills to be Assessed

This assignment assesses your skills in:

- Use of Power BI
- Use of different kinds of visualisations

Recommended Reading

The module notes are sufficient literature for completing this assignment successfully.

Equipment and Facilities to be Used

The assignment has to be implemented using Microsoft Power BI.

Workload

To successfully complete this assignment, a total of 20 hours should be budgeted.

Team Composition

This assignment is designed as a team project. Each team has the option to include two, three, or at most four members from the class. Feel free to choose your team members.

Working on group assignments helps develop crucial transferable skills like Teamwork, organization, communication, leadership, and problem-solving. Moreover, engaging in such collaborative work significantly contributes to your academic success and personal growth, especially since the capability to collaborate effectively is a vital prerequisite for numerous careers in the industry.

Task

The objective of this assignment is to create a Power BI dashboard that effectively visualises data from previous Summer and Winter Olympic Games. This dashboard will serve as a comprehensive tool to analyse and interpret various aspects of the Olympic data, providing insights into trends, performances, and other key metrics from past events.

Task Overview

1. **Data Collection**: Extract data from the Wikipedia page All-time Olympic Games medal table.
2. **Data Processing**: Clean and transform the data to make it suitable for analysis.
3. **Dashboard Development**: Create a Power BI dashboard that effectively visualises the data.
4. **Pattern Analysis**: Identify and describe key patterns and insights from the dashboard.

The dataset can be found at the following location:
https://en.wikipedia.org/wiki/All-time_Olympic_Games_medal_table

Instructions:

- Extract the tables containing the all-time medal counts for both Summer and Winter Olympics.
- Ensure the collection data on the following columns:

    - Country
    - Number of Summer/Winter Games participated
    - Gold medals
    - Silver medals
    - Bronze medals
    - Total medals
- Import the data into a suitable environment (e.g., Excel) for cleaning and transformation.
- Handle any missing values, inconsistencies, or formatting issues.
- Ensure that the data is structured in a way that Power BI can easily import and use it.
- Import the cleaned data into Power BI.
- Create a dashboard with multiple visualisations that may include:
    - Total medals won by each country in Summer vs. Winter Olympics.
    - Comparison of medal counts across different countries.
    - Trends over time for selected countries or overall.
    - Distribution of gold, silver, and bronze medals.
    - Participation vs. success rate (number of games participated vs. medals won).
- Ensure that the dashboard is interactive and user-friendly, with clear labels, titles, and legends.
- Analyse the visualisations to identify key patterns and insights.
- Look for trends such as:
    - Which countries dominate in the Summer vs. Winter Olympics?
    - How medal counts have changed over different Olympic periods.
    - Correlations between the number of participations and medals won.
- Document your findings in a report , describing the patterns you observed and any possible reasons behind them.

Remember, the key to this assignment is to extract meaningful insights from the data.

Sample Dashboard

Report

Submit a 3–5-page report in PDF format summarising the key patterns and insights you found from the dashboard.

Requirements / Marking Scheme

Requirement	Assessment Method	Weight (%)
Dashboard design	**Report/Power BI**	**70%**
Report	**Report**	**30%**

Notes

- The assignment must be completed with your team members. Any code snippets you or your team members do not directly write must be referenced within your code. You must directly comment on the code to explain its source. Failure to reference a code that is not yours will be treated as plagiarism.
- The assignment must be completed on time. If you submit work late, it will be marked according to the University's late submission policy

Submission

Your submission should be a single ZIP file uploaded via Teams. The file should be named as:

<<Your surname>>_<<Your name>>.zip — for example: **Smith_John.zip**.

All items in the zip file should also be prepared with your surname. (Ensure you replace "Smith" with your surname in the names below).

The following items must be included in your submission:

1. A folder named **Smith_Dashboard** contains the Power BI Dashboard.

2. A PDF document named **Smith_report.pdf** contains your report.

It is assumed that you will also address any social/legal and ethical issues surrounding the design and implementation of the project, such as copyright, references, and web law.